

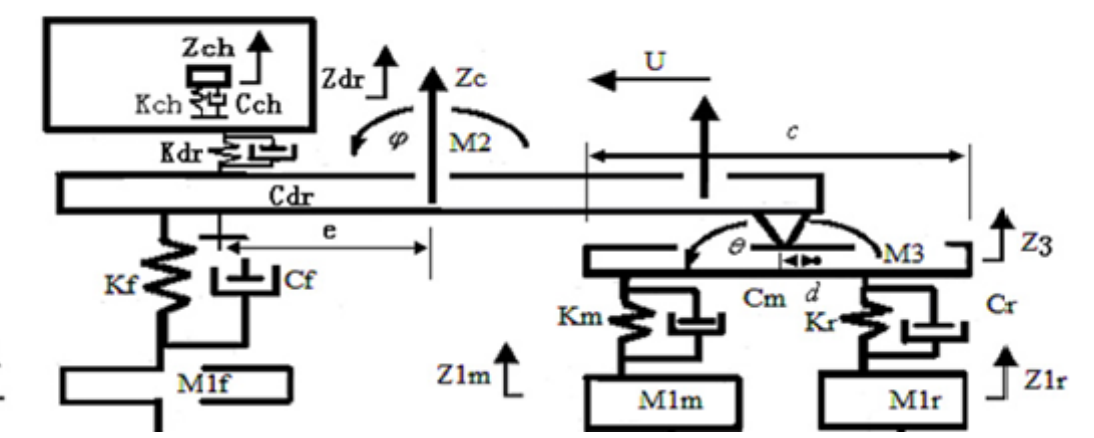
Physics-Informed Neural Network Based Surrogate Modeling for Ride Comfort Optimization of a 6×2 NRS Suspension System

PROBLEM STATEMENT

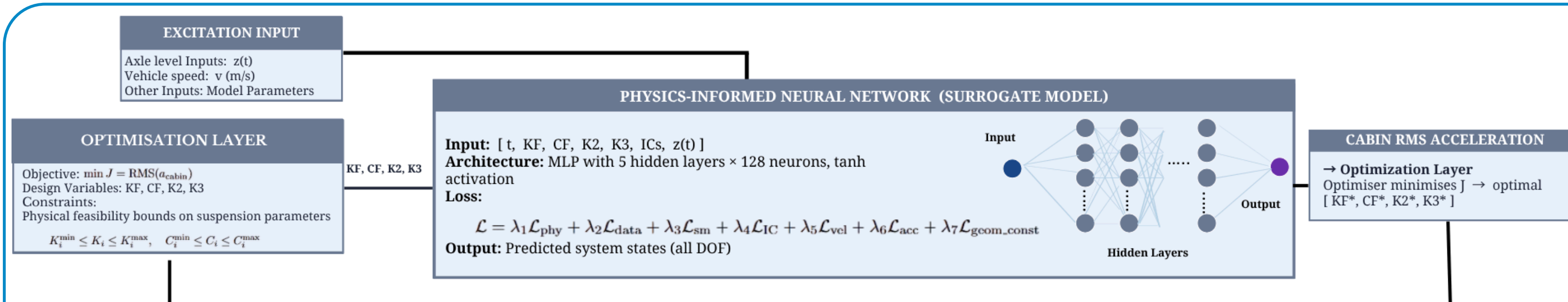
- Ride comfort is a key requirement in heavy vehicles, as cabin vibrations directly influence overall ride quality.
- Suspension systems in such vehicles are high-dimensional and strongly coupled, making their behaviour complex. The presence of geometric and mechanical constraints further adds to the difficulty in modelling and tuning.
- Conventional approaches rely on ODE-based simulations, which are computationally expensive for optimisation. Parameter tuning (such as stiffness and damping) requires repeated simulations, resulting in slow design cycles.
- This motivates the need for a fast, physics-consistent modelling to enable efficient suspension optimisation.
- In this work, a Physics-Informed Neural Network (PINN)-based surrogate model is proposed to address this challenge.

SYSTEM MODEL

- The system under consideration is a 6×2 heavy commercial vehicle equipped with a Non-Reactive Suspension (NRS) configuration, designed to improve ride comfort and stability.
- It consists of the cabin and chassis (sprung mass) with bounce (z), pitch (θ), and roll (ϕ) motions. Each of the three axes exhibits bounce (z) and roll (ϕ) dynamics.
- The rear tandem axles are mechanically coupled, which is a defining feature of the NRS setup. This arrangement allows load sharing without inducing additional pitch or roll moments, helping maintain vehicle attitude stability.



METHODOLOGY



Optimisation Framework

- The objective is to minimize cabin RMS acceleration
- A Bayesian optimization (BO) framework is employed, where the PINN acts as a fast surrogate of the underlying dynamics
- BO uses probabilistic acquisition function to guide sampling efficiently

PINN-Based Surrogate Model:

- PINN is trained to capture the system dynamics by combining multiple loss components, including data loss, physics loss (based on ODE residuals), constraint loss, smoothness, and losses on initial conditions, velocity, and acceleration.
- The network is an MLP, as specified, with a linear output layer
- The optimizer used is Adam for training
- The model is initially trained on synthetic data to ensure architecture correctness and stable learning

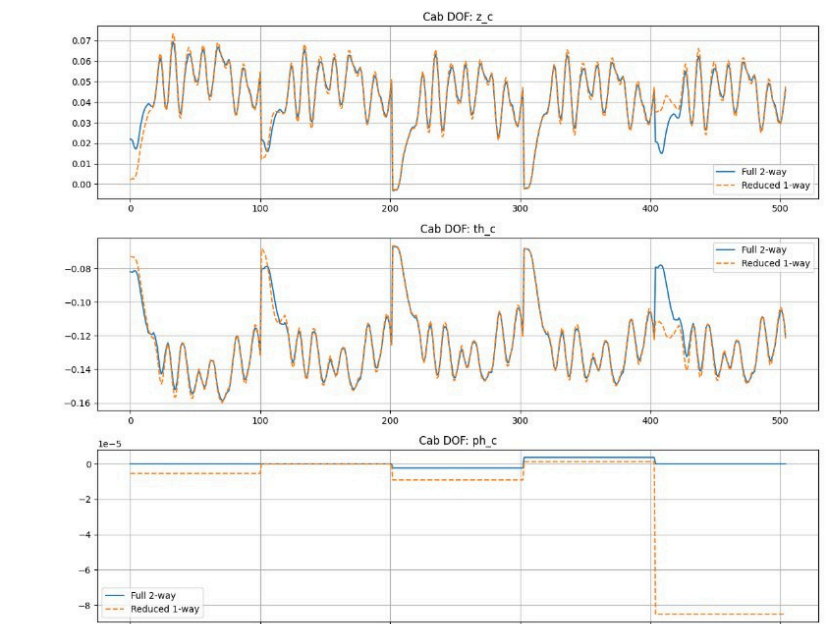
Data Generation

- Synthetic data is generated by solving a 12-DOF suspension model with two geometric constraints
- The system is formulated using KKT conditions and stabilized through Baumgarte stabilization to ensure constraint consistency

$$M(q) \ddot{q} + R(q, \dot{q}, t) + G(q)^T \lambda = 0$$

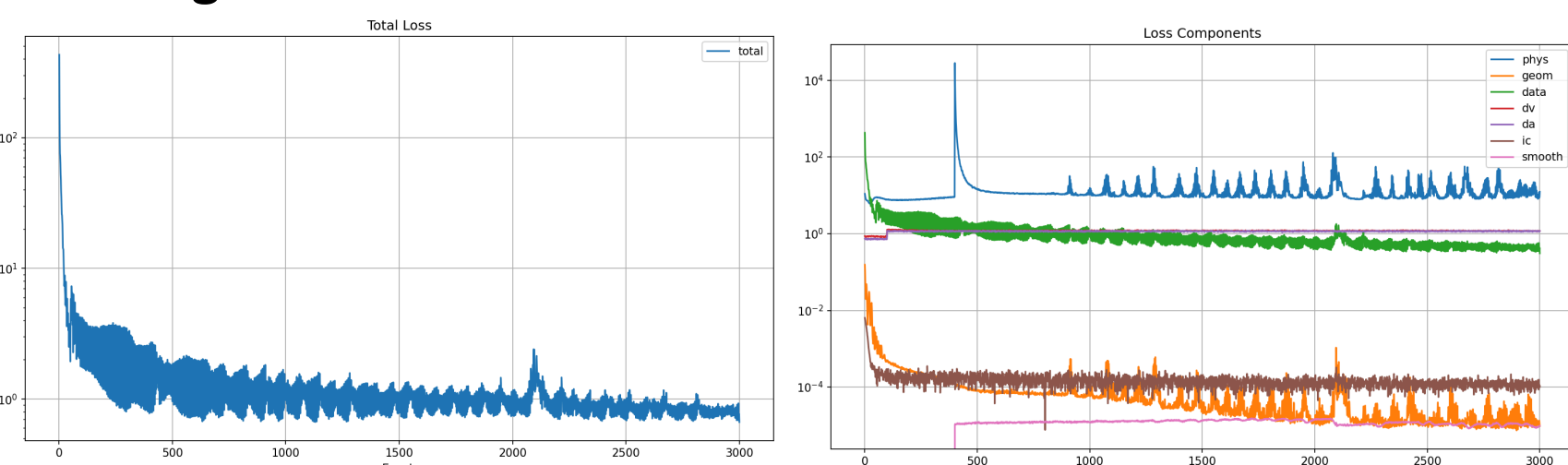
$$\gamma = 2\zeta\omega \dot{q}(q) + \omega^2 g(q) \quad G(q) \ddot{q} = -\gamma$$

$$\begin{bmatrix} M & G^T \\ G & 0 \end{bmatrix} \begin{bmatrix} \ddot{q} \\ \lambda \end{bmatrix} = \begin{bmatrix} -R \\ -\gamma \end{bmatrix}$$



RESULTS

Training Results:



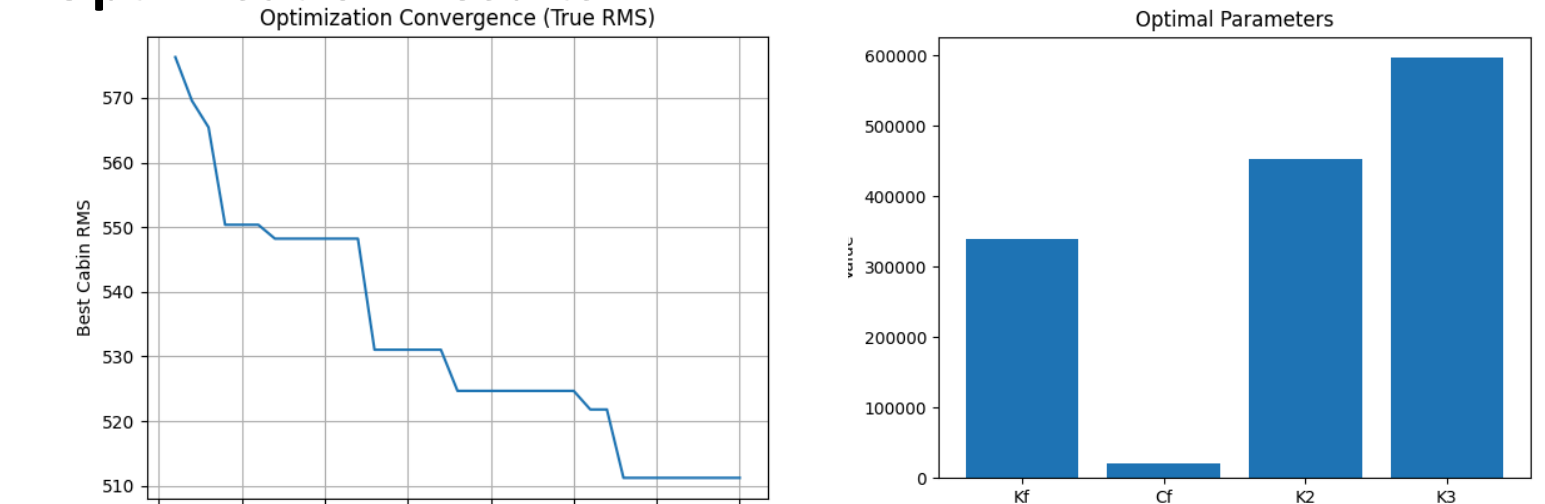
- Model trained for 3000 epochs with stable convergence
- Total loss shows rapid initial decay followed by smooth stabilization

Test Results:

Metric	Value
RMSE	7.28×10^{-3}
MAE	4.93×10^{-3}
R^2	0.716
Accuracy	72.1%

- Model shows good generalization on unseen data

Optimisation Results:



- Achieved 12.6% cabin RMS reduction in 35 iterations, confirming convergence, while Cf shows minimal impact vs. Kf, K2, K3

Automated Engineering Drawing Compliance Checker

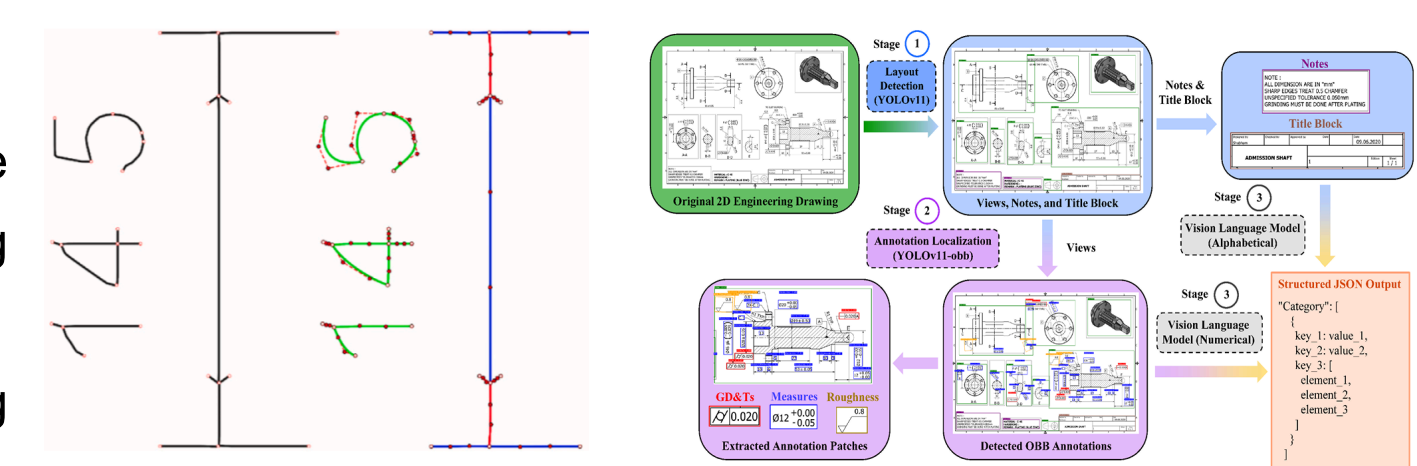
METHODOLOGY & RESULTS

Problem Setup:

- Engineering drawings must comply with strict standards to ensure manufacturability and correctness.
- Automated Checker Framework:
- Input Processing: Extract geometric features and annotations (PDF/DXF)
- Rule-Based Engine: Encode checklist rules, map features, validate compliance
- Feedback Generation: Pass / Fail output, Flags: missing annotations, incorrect dimensions, standards violations

Methods explored and Results:

- Input Processing: Engineering drawings (PDF/DXF) are processed via three pipelines image-based (YOLOv8), OCR-based, and direct CAD parsing (ezdxf).
- YOLOv8 Pipeline: View & annotation detection: preprocessing (skeletonization, stroke extraction, Bezier fitting) for reconstruction
- Raster Limitations: Fidelity loss, line/arc ambiguity, noise sensitivity: unreliable reconstruction
- OCR Parsing: Extracts text/annotations; layout inferred: weak view separation & geometry understanding
- DXF Extraction: Precise primitives (lines, arcs, circles, polylines) from native vectors: no raster loss, but here there are associativity issues.



Conclusion:

- The system is limited by the absence of a unified approach that ensures both reliable geometric extraction and contextual understanding across widely varied engineering drawings, leading to inconsistent performance and hindering end-to-end automation.

SUMMARY & FUTURE EXTENSION

- The modular framework enables easy integration of alternative optimization strategies, shows reliable performance on synthetic data, and requires further testing across optimization methods alongside transition to real-world deployment.
- Extend the model to include shock absorber dynamics and validate under realistic conditions, while optimizing hyperparameters of PINN to improve accuracy and reliability.
- Integrate the model into a unified agentic framework, enabling intelligent redirection for suspension design queries, while finalizing and developing the overarching system for seamless end-to-end deployment.